

and can cover a capacitance range from 47nF to 1µF.

In contrast to conventional MLCCs, these flip-type capacitors have connec-

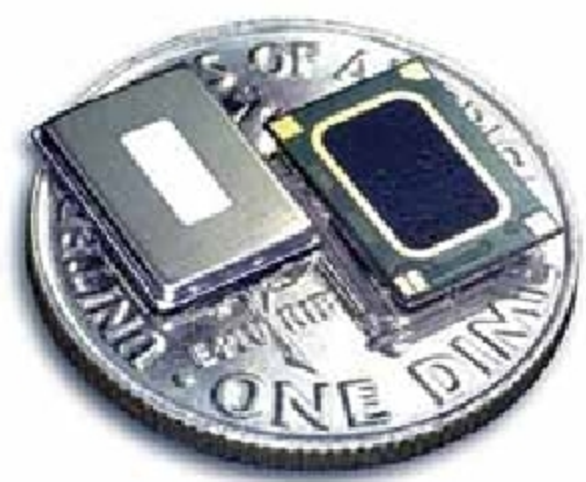


tions that can be rotated through 90 degrees, producing a wider and shorter current path, which results in lower equivalent series inductance (ESL) and equivalent series resistance (ESR) values, and thus lower impedances.

TDK Corporation
www.tdk.com

Water and dust-resistant micro-speaker

A new MEMS speaker called Montara has been recently launched. As per the product's press release, it can deliver



high-fidelity, full-bandwidth sound and low total harmonic distortion (THD) for sealed in-ear personal audio devices, including true wireless stereo (TWS) earbuds. Montara is also an IP-57 rated micro-speaker, enabling water and dust-resistant earbuds without special protective mesh or coverings.

Consumers can benefit from enhanced audio fidelity with raised levels of clarity and detail for instruments and vocals, extremely low latency, waterproof earbuds, and longer listening times.

Montara replaces the previous highly-variable voice-coil speaker technology with a high-precision, high consistency, and monolithic semiconductor process

by implementing the entire speaker (actuator and diaphragm/membrane) in silicon, thus reducing package height and eliminating driver matching and calibration due to inherent variability in membrane assembly.

xMEMS Labs
www.xmems.com

Wireless charging for robots and UVs

As robotic fleets reshape logistics, delivery, and inspection industries, the demand for more efficient and flexible



charging solutions is increasing while the feasibility for these fleets to be managed manually and recharged by a 24/7 rotation of personnel is decreasing. This calls for a step towards enhanced autonomy, whereby normal operations can be sustained without human intervention. This is where WiBotic comes in.

WiBotic is a provider of wireless charging and power optimisation solutions that are integral to a rapidly growing ecosystem of aerial, mobile, marine, and industrial robots. Its range of solutions enable robots and unmanned vehicles (UVs) to be recharged through wireless charging stations.

Vicor's Zero-Voltage Switching (ZVS) regulator is built for powering wireless charging. The Vicor 48V VI Chip PRM Regulator is a 400W high-efficiency converter that operates from a 36 to 75V input to generate a regulated output. The device powers the adaptive matching transmitter onboard the WiBotic TR-110 wireless charging station, which feeds power wirelessly to the robot's or UV's onboard receiver.

Vicor
www.vicorpower.com

For queries or submissions regarding Truly Innovative Tech, contact us at innov8@efy.in

Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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